Henry Yip

MSc Physics student, ETH Zürich

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Research interests: tropical cyclones, numerical weather prediction, climate modelling, and machine learning for atmospheric science.

Education

MSc Physics, ETH Zürich — Sep 2025 – Aug 2027 (expected)

Current average 5.42 out of 6. Courses include Computational Statistical Physics (6.0, full marks); Cloud Dynamics: Hurricanes (5.75); Dynamics of the Large-Scale Atmosphere; Computational Astrophysics; AI in the Sciences and Engineering; Solid State Theory; Advanced Solid State Physics; Ultrafast Laser Physics.

BSc (Hons) Mathematical Physics, University of Edinburgh — Sep 2021 – May 2025

First Class Honours, four-year average 76%. Courses include Numerical ODEs (92%), Modelling and Visualisation in Physics (84%), Methods of Mathematical Physics (82%), Computer Modelling (81%) and Thermal Physics (81%).

Publications

1. **H. Yip** et al., *A Model-Agnostic Post-Processor for Tropical Cyclone Intensity and Central Pressure in Machine-Learning Weather Forecasts*. Submitted to Tropical Cyclone Research and Review (2026).
2. **H. Yip** and J. M. Smillie, *An Analysis of the Restricted Euler Problem Using Symplectic Integrators*. Edinburgh Student Journal of Science **1**(3) (2025), peer-reviewed. doi:10.2218/esjs.10064.

Research experience

Master's thesis, Atmospheric Physics group, ETH Zürich — Feb – Aug 2027 (planned)

- *Incorporating extratropical transition into the ICON tropical cyclone tracker.* Will compare definitions of extratropical transition based on the cyclone phase space on the tracks of the multi-threshold ICON tracker of Enz et al. (2023), find at which stage of transition the tracks end, continue tracks that stop too early, and measure how much accumulated cyclone energy comes from after the transition.
- Supervised by Prof. Ulrike Lohmann and Dr Bernhard Enz.

Summer intern, Hong Kong Observatory — Jun – Sep 2026

- Developed a model-agnostic machine-learning post-processor for tropical cyclone intensity and central-pressure forecasts from global AI weather models, training convolutional neural networks and Transformers in PyTorch. First-author manuscript submitted (publication 1).

Semester project, Atmospheric Physics group, ETH Zürich — Feb – Jul 2026

- The resolution dependence of environmental tropical cyclone metrics: how coarsening ERA5 ensemble reanalysis from its native 0.5° grid to 0.75°–2.0°, and averaging it over 6 h to 30 days, changes tropical cyclone potential intensity over the western tropical Atlantic (ten members; July–October of 2005, 2010, 2016 and 2018).
- Differences between grids concentrate near the coast. Excluding peaks within 100 km of it brings the monthly mean mismatch between the strongest peaks on the 0.5° and 0.75° grids below $2\,\mathrm{m\,s^{-1}}$ in 11 of 16 month–year cases, from 2.9–11.1 $\mathrm{m\,s^{-1}}$ unfiltered. Monthly averaging lowers peak potential intensity by 8%.
- Built the data pipelines in Python (xarray, netCDF4, NumPy) and CDO on the CSCS Santis supercomputer: ensemble downloads, remapping, and parallel computation with SLURM.
- Supervised by Prof. Ulrike Lohmann and Dr Bernhard Enz.

Final project, Computational Statistical Physics, ETH Zürich — Spring 2026

- Monte Carlo study of the 3D classical Heisenberg model with Metropolis and Wolff cluster updates on lattices up to 256^3 spins: the critical temperature and five critical exponents, all within 1σ of the literature values. Write-up.

Synchronisation in 2D turbulence, School of Mathematics, University of Edinburgh — Jun – Sep 2025

- Analysed small-scale synchronisation in two-dimensional Kolmogorov-forced turbulence with JAX-CFD: implemented a splitting of the spectral solver into large- and small-scale modes, measured exponential synchronisation rates across Reynolds numbers, and began extending the analysis to Coriolis effects on the β-plane.
- Supervised by Dr Jacob Page, with assistance from Dr Dmitriy Zhigunov.

The three-body problem, School of Physics and Astronomy, University of Edinburgh Jun – Sep 2024

- Implemented and benchmarked symplectic integrators (Verlet, Ruth, Neri) and verified known periodic orbits (figure-eight, Lagrange, Broucke–Hadjidemetriou–Hénon) and Euler's problem of two fixed centres against published results (publication 2).
- Worked through Chenciner and Montgomery's variational proof that the figure-eight orbit exists; studied the orbit's stability under perturbations and searched for new well-behaved orbits.
- Supervised by Prof. Jenni Smillie; funded by a Career Development Summer Scholarship. Presented at the university's summer project awards ceremony. Code and write-up.

Group leader, liquid slosh project, Edinburgh Scientific Researchers Association Sep 2023 – Mar 2024

- Modelled liquid slosh with coupled pendulums to capture the motion of the surface, derived the equations of motion, and formulated an alternative model with the wave equation. Presented at the association's annual conference.

Teaching

- Peer tutor with the Math Buddies scheme, University of Edinburgh: tutored students in first-year mathematics and physics courses, and received the Edinburgh Award (2025) for this work.

Awards

- Edinburgh Award (2025), for tutoring with the Math Buddies scheme.
- Career Development Summer Scholarship (2024), £2,520, awarded to top students for physics summer research projects.
- Pre-Honours Certificate of Merit (2022 and 2023), for first-class performance in years 1 and 2.

Leadership and outreach

- Programme Representative, Edinburgh University Students' Association (2021–25): represented students to the visiting Institute of Physics panel and in Assessment and Feedback workshops with lecturers.
- Hyperloop Edinburgh (HYPED): research analyst on liquid ring pumps for hyperloop vacuum systems (2022–23) and outreach coordinator (2021–22).
- Mathematics and physics blog (2022–present): over 35 posts on classical mechanics, number theory, numerical methods, Python and LaTeX, including a week-by-week research log of the 2024 three-body project.

Skills

Programming	Python (PyTorch, NumPy, SciPy, Matplotlib, xarray, netCDF4, JAX, JAX-CFD, Numba)
Machine learning	Convolutional neural networks and Transformers (PyTorch)
Computing	SLURM and high-performance computing (CSCS Santis, ETH Euler), CDO, ERA5 reanalysis
Tools	Git and GitHub, LaTeX, Jekyll
Languages	English, Cantonese and Mandarin (fluent); French (basic, IGCSE grade A); German (basic)